Literature resolution of the Wang–Madiman Rényi EPI conjecture

Source: arXiv:1604.04225   Answer: arXiv:1904.08038

Status. Literature already answered; no new mathematical result is claimed here.

Source question. In Section 2.3.3, pp. 14–16, of Madiman, Melbourne, and Xu, *Forward and Reverse Entropy Power Inequalities in Convex Geometry*, the Wang–Madiman conjecture is stated as follows. Let $X_1, \dots, X_n$ be independent random vectors in $\mathbb{R}^d$, let $p > d/(d+2)$, and choose independent scaled generalized Gaussians Z_i so that $h_p(X_i) = h_p(Z_i)$. Then

$$N_p(X_1 + \cdots + X_n) \geq N_p(Z_1 + \cdots + Z_n).$$

The survey reports that all cases other than the classical $p = 1$ case and the established $p = \infty$ cases remained open.

Exact later answer. Jaye, Livshyts, Paouris, and Pivovarov, *Remarks on the Rényi Entropy of a sum of IID random variables*, arXiv:1904.08038, restate this assertion as Conjecture 1.1. Their Introduction states that it fails already when

$$d = 1, \qquad p = 2, \qquad n = 2,$$

with X_1, X_2 independent and identically distributed. Section 4 applies the Euler–Lagrange equation derived in Section 3 and shows that the proposed generalized Gaussian does not satisfy the necessary first-order condition. Consequently it is not a minimizer of the Rényi entropy of the sum under the fixed-entropy constraint, disproving the conjecture.

Scope. The conjectured extremizer is decisively ruled out, but the answering paper leaves the identity of the true minimizer open.